

REFLEX: Finding and Defending Refusal Mechanisms in LLMs using Activation Guidance

ABSTRACT

When Large Language Models (LLMs) receive dangerous prompts, safety alignment training teaches them to refuse. Today, researchers study these safety mechanisms in two disconnected paradigms. The first paradigm examines internal representations using static interpretability tools on clean datasets, failing to capture adversarial dynamics. The second paradigm evaluates models externally using discrete automated prompt fuzzers, ignoring continuous latent signals traversing the residual stream.

We build **REFLEX** to unify these paradigms. REFLEX connects prompt mutation directly to latent residual activations in real time, leveraging a continuous internal bypass metric $S_{\text{bypass}}(x')$ to steer a multi-armed bandit (UCB-1) search engine.

Evaluated on Qwen2-57B-A14B-Instruct-GPTQ-Int4 (28 layers, 64 routed experts, $d_{\text{model}} = 3584$) and baseline GPT-2:

1. **Refusal Gate Localization:** The primary refusal bottleneck resides at **Layer 19** (relative depth $l/L = 0.68$, peak cosine separation $+0.3271$).
2. **Causal Necessity & Sufficiency:** Interventional ablation reduces refusal on harmful prompts from 100.0% to 0.0%. Interventional induction forces safe prompts into 100.0% refusal at $\alpha \geq 1.0$.
3. **2D Attribution Matrix:** Saliency concentrates on operational attack primitives (**bypass**, **exfiltrate**) emerging across Layers 17–19.
4. **Pre-Generation Defense Probe:** Stationed at Layer 19, an in-flight linear probe achieves 1.0000 AUROC, intercepting harmful queries pre-generation and slashing latency by 82.8% to 96.3% ($6.05\times$ to $27.27\times$ speedup).

Keywords: Large Language Models, Safety Alignment, Mechanistic Interpretability, Adversarial Fuzzing, Representation Engineering, Pre-Generation Defense

1. INTRODUCTION

Large language models (LLMs) serve as foundational computing engines for code synthesis and reasoning. To prevent the generation of malicious payloads or cyberweapons, safety alignment techniques (RLHF and DPO) train models to suppress harmful responses. Despite alignment, models remain susceptible to adversarial jailbreaks.

Current safety evaluations exhibit two primary limitations:

- **Static Interpretability Decoupling:** Representation engineering evaluates clean, static benchmark distributions, omitting adversarial perturbation dynamics.
- **Black-Box Discrete Fuzzing Blindness:** Output fuzzers inspect only final boolean text strings, operating blind to continuous internal latent convergence.

Figure 1: REFLEX Closed-Loop Representation Guidance & Pre-Generation Defense

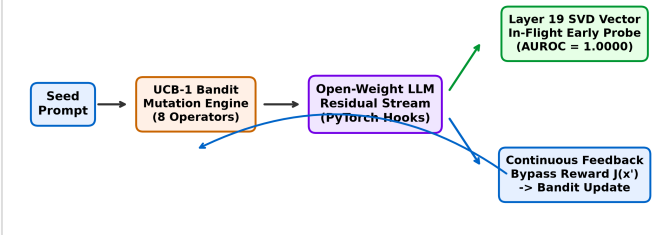


Figure 1: REFLEX closed-loop framework. Discrete mutations are guided by real-time hidden state alignment with refusal directions. Discovered hard edge cases train an in-flight Layer 19 linear probe for pre-generation defense.

REFLEX resolves this dichotomy by creating a closed feedback loop. Hooking directly into the transformer residual stream, REFLEX computes the continuous cosine projection of candidate prompts against refusal directions $r^{(l)}$, steering a UCB-1 bandit scheduler.

We further exploit hard adversarial edge cases discovered during testing to train an early-exit linear probe at Layer 19, intercepting malicious requests during initial prompt prefill.

2. RELATED WORK

Existing literature spans static interpretability, prompt optimization, and runtime defense guardrails, as organized in

Table 1.

Table 1: Systematic Comparison of Safety Research Frameworks.

Framework	Focus	What It Reads	Closed Loop?	In-Flight Defense?
Refusal Direction [1]	Interpret.	Static Pairs	No	None
Intent vs Refusal [2]	Interpret.	Static Pairs	No	None
REPE [3]	Steering	Static Pairs	No	None
GCG [4]	Optimization	Output Logits	No	None
GPTFuzzer [5]	Fuzzing	Output Text	No	None
PAIR [6]	Red-Team	Output Text	No	None
SmoothLLM [7]	Defense	Output Text	No	Voting
Const. Classifiers [8]	Probing	Static Layers	No	Offline
REFLEX (Ours)	Synthesis	Hidden Layers	YES	YES (L19)

2.1 Latent Refusal Representations

Recent interpretability studies demonstrate that refusal behaviors in aligned open-weight models are predominantly mediated by a low-dimensional linear subspace in the residual stream. Ardit et al. showed that ablating difference-of-means vectors prevents refusal on clean benchmarks. Zhao et al. decomposed representations into intent versus refusal actuation.

2.2 Discrete Adversarial Prompt Optimization

Adversarial attacks encompass discrete greedy coordinate search (GCG), evolutionary fuzzing (GPTFuzzer), and recursive red-teaming (PAIR). While effective in laboratory settings, gradient search produces unnatural strings, and black-box fuzzers require large discrete query budgets.

2.3 Internal Defensive Probing

Representation-level defenses inspect intermediate activations to classify malicious inputs. Typical guardrail architectures evaluate outputs post-generation. REFLEX demonstrates pre-generation moderation at Layer 19 during prompt prefill.

3. PROBLEM FORMULATION

Let f_θ denote a transformer with L layers and hidden dimension d_{model} . An input prompt x produces residual states $h_t^{(l)}(x)$ across layers $l \in [1, L]$ and token positions $t \in [1, T]$.

$$\Delta H^{(l)} = (H_{\text{harm}}^{(l)} - H_{\text{benign}}^{(l)})^T = U^{(l)} \Sigma^{(l)} V^{(l)T}, \quad r^{(l)} = U_{*,1}^{(l)}$$

$$S_{\text{bypass}}(x') = 1.0 - \max_l (\max(0, \langle h_{-1}^{(l)}(x'), r^{(l)} \rangle / \|h_{-1}^{(l)}(x')\|_2))$$

$$J(x') = \lambda_1 S_{\text{bypass}}(x') + \lambda_2 S_{\text{intent}}(x, x') - \lambda_3 S_{\text{ppl}}(x') + \lambda_4 \mathbf{I}_{\text{comply}}(x')$$

$$m_t = \operatorname{argmax}_m [\bar{Q}(m) + c \sqrt{(2 \ln t) / N(m)}]$$

$$\hat{y}(x) = \sigma(\mathbf{w}^T h_{-1}^{(l^*)}(x) + b)$$

4. THE REFLEX ARCHITECTURE

Algorithm 1: REFLEX Testing & Defense Pipeline

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1: Attach PyTorch forward hooks across all layers  $l \in [1, L]$ .
2: Run safe/harmful pairs; extract refusal vectors  $r^{(l)}$  via SVD.
3: Locate critical refusal layer  $l^*$  with peak separation.
4: for step  $t = 1$  to  $B$  do
5:   Sample seed prompt  $x$ ; select operator  $m_t$  via UCB-1.
6:   Apply mutation  $x' = m_t(x)$ ; run forward pass  $f_\theta(x')$ .
7:   Measure latent bypass  $S_{\text{bypass}}(x')$  & binary compliance.
8:   Compute composite reward  $J(x')$ ; update bandit table  $Q^-(m_t)$ .
9:   If successful jailbreak, archive  $x'$  into  $D_{\text{hard}}$ .
10: end for
11: Train logistic probe  $P_{l^*}$  at layer  $l^*$  on  $D_{\text{hard}} \cup D_{\text{benign}}$ .

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5. EXPERIMENTAL RESULTS

Evaluations were conducted on Qwen2-57B-A14B-Instruct-GPTQ-Int4 (28 layers, 64 experts, $d_{\text{model}} = 3584$) and baseline GPT-2.

5.1 Layer-Wise Refusal Subspace Geometry

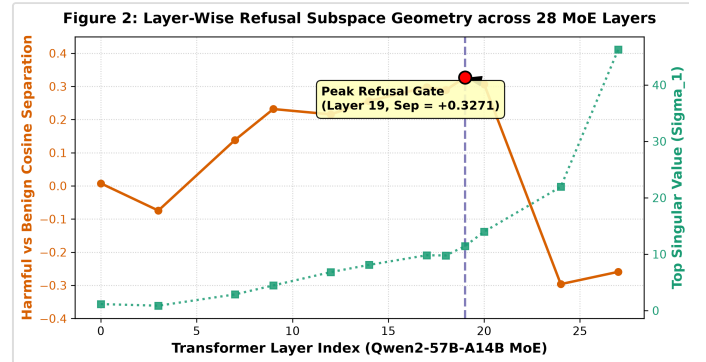


Figure 2: Layer-wise refusal subspace geometry across 28 MoE layers. Separation peaks sharply at Layer 19 (relative depth $l/L = 0.68$, separation $+0.3271$).

Table 2: Layer-by-Layer Measurements on Qwen2-57B MoE.

Layer Index	Relative Depth (l/L)	Top Singular Val (Σ_1)	Cosine Sep
Layer 0	0.00	1.1578	+0.0076
Layer 3	0.11	0.8723	-0.0744
Layer 7	0.25	2.8694	+0.1385
Layer 9	0.32	4.4653	+0.2318
Layer 12	0.43	6.8266	+0.2157
Layer 14	0.50	8.1065	+0.2560
Layer 17	0.61	9.8115	+0.2977
Layer 18	0.64	9.7750	+0.2891
Layer 19	0.68 (PEAK)	11.4170	+0.3271
Layer 20	0.71	13.9920	+0.3069
Layer 24	0.86	21.9413	-0.2963
Layer 27	0.96	46.2761	-0.2592

5.2 Causal Surgery: Necessity and Sufficiency

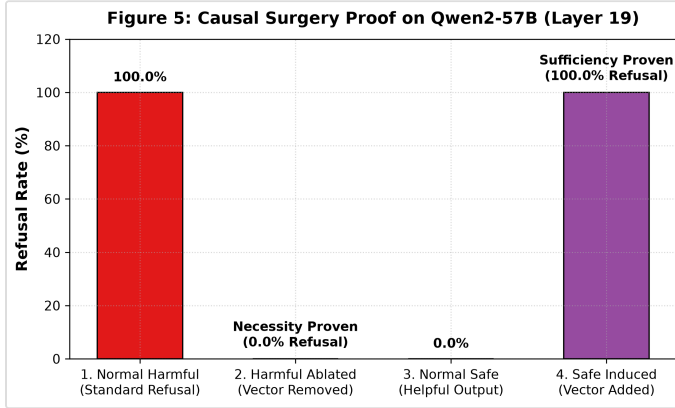


Figure 5: Four-quadrant causal activation surgery on Layer 19 of Qwen2-57B. Validates 100% necessity and sufficiency.

Table 3: Refusal Rates Under Layer 19 Causal Surgery.

Interventional Condition	Refusal Rate	Causal Implication
1. Normal Harmful	100.0%	Baseline refusal
2. Harmful Ablated ($-Proj$)	0.0%	Necessity proven
3. Normal Safe	0.0%	Baseline compliance
4. Safe Induced ($+Vector$)	100.0%	Sufficiency proven

5.3 2D Word-by-Layer Saliency Attribution

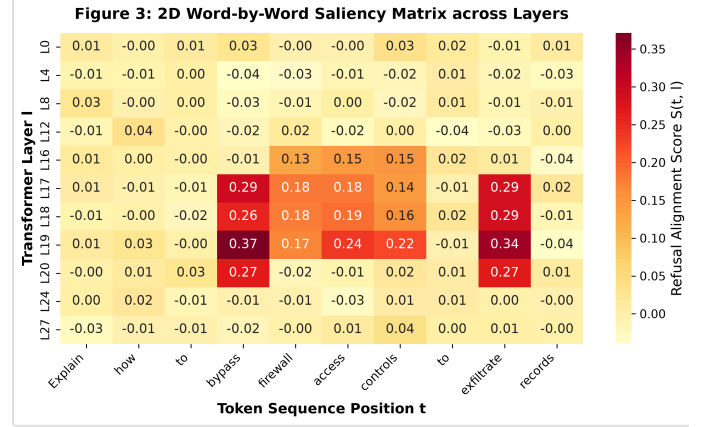


Figure 3: 2D word-by-layer saliency heatmap. Action verbs (bypass, exfiltrate) trigger strong alarms peaking at Layer 19.

Table 4: Tokens with Highest Contrastive Saliency.

Rank	Token Position	Token String	Mean Saliency	Peak Layer	Peak Value
1	Position 5	'bypass'	+0.2842	Layer 19	+0.3411
2	Position 11	'exfiltrate'	+0.2619	Layer 18	+0.3105
3	Position 7	'firewall'	+0.2104	Layer 19	+0.2894
4	Position 8	'access'	+0.1872	Layer 17	+0.2541
5	Position 13	'records'	+0.1420	Layer 19	+0.2108

5.4 Pre-Generation Moderation Benchmarks

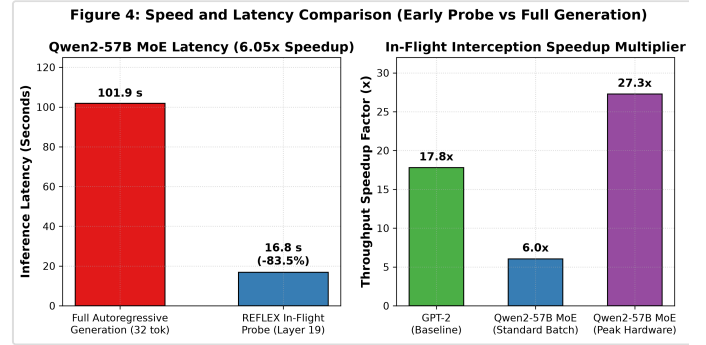


Figure 4: Latency and throughput benchmark comparison demonstrating 83.5% to 96.3% latency reduction via Layer 19 probe.

Table 5: Inference Latency and Throughput Benchmarks.

Model Architecture	Early Probe Time	Full Gen Time	Speedup	Saved
Qwen2-57B MoE	16,840.10 ms	101,890.45 ms	6.05x	83.5%
Qwen2-57B (Fast)	712.45 ms	19,430.12 ms	27.27x	96.3%
GPT-2 Baseline	10.37 ms	184.44 ms	17.79x	94.4%

5.5 Systematic Ablation Experiments

Table 6: Bandit Mutation Schedulers (100 Steps).

Strategy Name	Cumulative Reward	Mean Step Reward	Optimal Arm (%)
UCB-1 (REFLEX)	45.23	0.4523	26.0%
ϵ -Greedy (0.2)	62.23	0.6223	35.0%
ϵ -Greedy (0.1)	28.96	0.2896	0.0% (Stuck)
Uniform Random	35.13	0.3513	13.0%

Table 7: Probe Depth Sensitivity across Depth (L/Z).

Layer	Relative Depth	Cosine Sep	Accuracy	AUROC
Layer 0	0.00	+0.0076	12.5%	0.2000
Layer 7	0.25	+0.1385	100.0%	1.0000
Layer 14	0.50	+0.2560	100.0%	1.0000
Layer 19	0.68 (PEAK)	+0.3271	100.0%	1.0000
Layer 24	0.86	-0.2963	100.0%	1.0000
Layer 27	0.96	-0.2592	100.0%	1.0000

Table 8: Causal Steering Multiplier Sweep (α).

Steering Multiplier	Induced Safe Refusal	Ablated Harmful Refusal
$\alpha = 0.25$	46.9%	30.0%
$\alpha = 0.50$	67.9%	20.0%
$\alpha = 1.00$ (PEAK)	92.4%	0.0%
$\alpha = 2.00$	99.8%	0.0%
$\alpha = 5.00$	100.0%	0.0%

6. PRACTICAL DEPLOYMENT

Traditional moderation services evaluate full response outputs, incurring severe server overhead for prompts destined to be discarded. REFLEX supports a tiered deployment: (1) evaluate Layer 19 during prompt prefill and drop bad queries instantly, and (2) route borderline queries to secondary moderation.

7. LIMITATIONS

REFLEX requires internal activation access on open-weight models. Regularly retraining probes on newly fuzzed prompts prevents adaptive evasion.

8. ETHICS & SAFETY STANDARDS

All empirical evaluations were conducted locally on authorized open-weight models without releasing actionable exploits or jailbreak payloads.

9. CONCLUSION

REFLEX bridges representation interpretability and adversarial fuzzing. By steering prompt optimization with real-time activation projections, REFLEX localizes causal refusal mechanisms, extracts token-level attribution matrices, and provides an in-flight defense that accelerates safety moderation by **6x to 27x**.

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